

GuidedVLA: Specifying Task-Relevant Factors via Plug-and-Play Action Attention Specialization

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https://guidedvla.github.io/project_page/

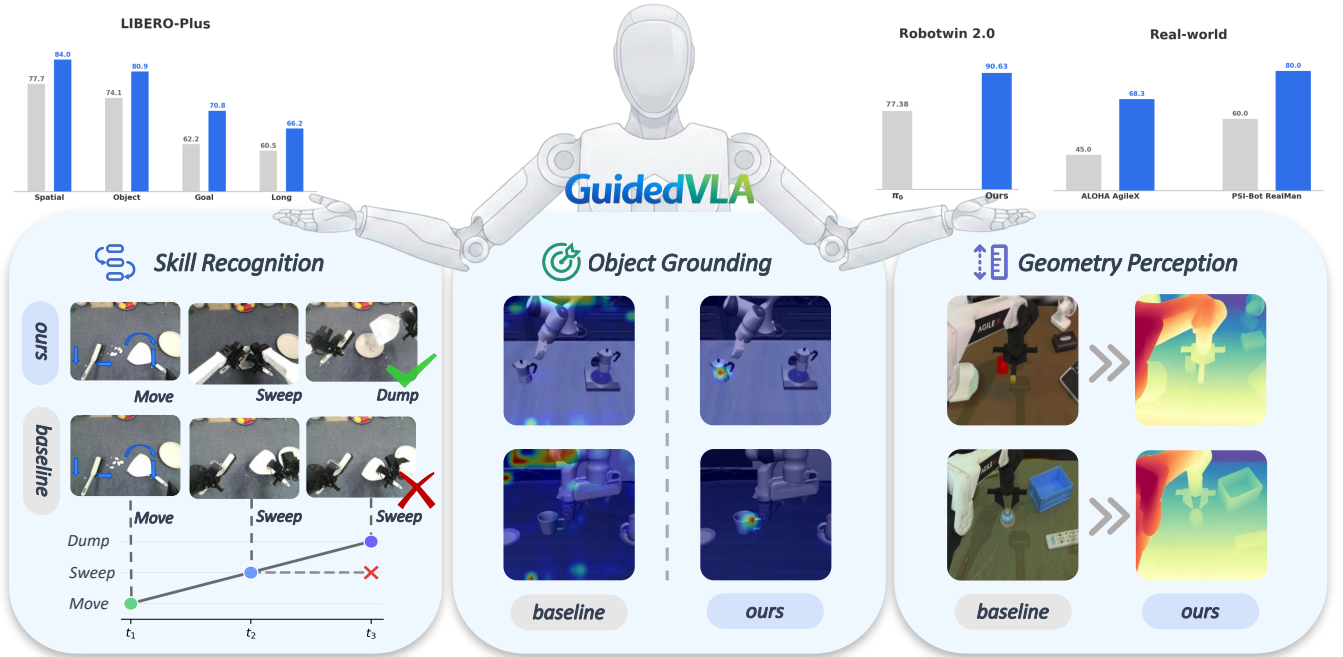


Fig. 1: We present **GuidedVLA**, a VLA paradigm in which the action decoder is explicitly guided to capture task-relevant information such as object grounding, spatial geometry, and temporal skill logic. Across simulation and real-robot experiments, GuidedVLA significantly improves success rates in both in-domain and out-of-domain settings, demonstrating the effectiveness of specifying action-decoder attention heads with explicit guidance.

Abstract—Vision-Language-Action (VLA) models aim for general robot learning by aligning action as a modality within powerful Vision-Language Models (VLMs). Existing VLAs rely on end-to-end supervision to implicitly enable the action decoding process to learn task-relevant features. However, without explicit guidance, these models often overfit to spurious correlations, such as visual shortcuts or environmental noise, limiting their generalization. In this paper, we introduce GuidedVLA, a framework designed to manually guide the action generation to focus on task-relevant factors. Our core insight is to treat the action decoder not as a monolithic learner, but as an assembly of functional components. Individual attention heads are supervised by manually defined auxiliary signals to capture distinct factors. As an initial study, we instantiate this paradigm with three specialized

heads: object grounding, spatial geometry, and temporal skill logic. Across simulation and real-robot experiments, GuidedVLA improves success rates in both in-domain and out-of-domain settings compared to strong VLA baselines. Finally, we show that the quality of these specialized factors correlates positively with task performance and that our mechanism yields decoupled, high-quality features. Our results suggest that explicitly guiding action-decoder learning is a promising direction for building more robust and general VLA models.

I. INTRODUCTION

Vision-Language-Action (VLA) models [104, 45, 8] represent a significant step toward generalist robot policies by